

Week 0 (CS 418 @ UIC)

Week 0 Slide Deck

An Experiment With Quarto

Jack Bandy 2026

This is a demo slide deck for reference and experimentation.

Harold Washington Library

CS 418 · Week 0 ● Harold Washington Library ●





Content Slide

Testing content slide.

- Lorem ipsum dolor sit amet
- Consectetur adipiscing elit
- Sed do eiusmod tempor incididunt ut labore et dolore magna aliqua
- Test quote

“The greatest value of a picture is when it forces us to notice what we never expected to see.” — John Tukey

Incremental Bullets

- display this first
- display this second
- display this third
- display this fourth
- display this fifth

Tabsets

Drop a finger on a random spot of the spinning globe: **Water** or **Land**? About 75% of Earth's surface is water, so each toss is a coin flip weighted by `p`.

Python

R

Julia

JavaScript

```
1 import random
2
3 def globe_toss(p=0.75):
4     return "Water" if random.random() < p else "Land"
```

Pretty Code

- Syntax highlighting works in revealjs slides
- Code uses the theme's Courier stack

```
1 import polars as pl
2
3
4 def summarize_ridership(stations: pl.DataFrame) -> pl.DataFrame:
5     return (
6         stations
7         .group_by("line")
8         .len(name="station_count")
9         .sort("station_count", descending=True)
10    )
```

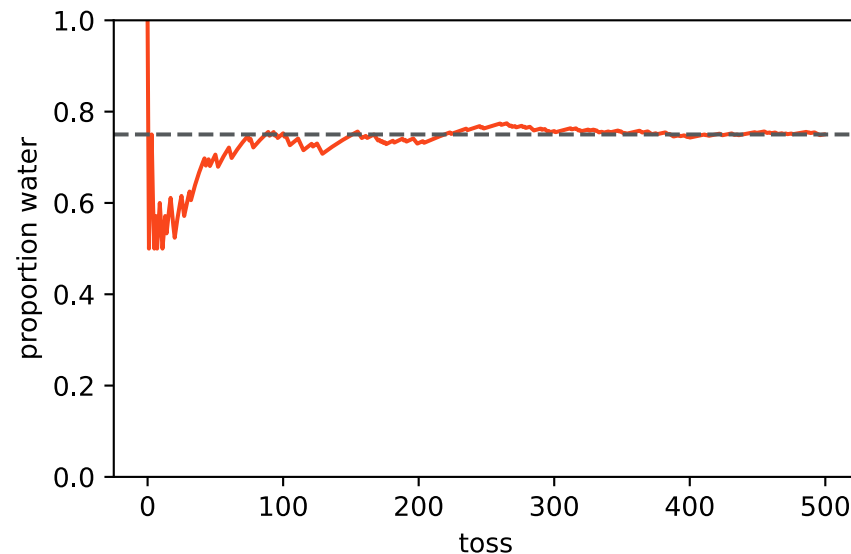

Line Highlighting

```
1 import numpy as np
2 import polars as pl
3
4 stations = pl.read_csv("cta_stations.csv")
5 orange = stations.filter(pl.col("line") == "Orange")
6
7 by_access = orange.group_by("accessible").len(name="station_count")
8 share = np.round(
9     by_access["station_count"] / by_access["station_count"].sum(),
10     2,
11 )
12
13 print(by_access.with_columns(accessible_share=share))
```

Code + Figure

The globe toss from earlier: the proportion of water converges on p .

```
1 import numpy as np
2 import matplotlib.pyplot as plt
3 water = np.random.default_rng(418).random(500) < 0.75
4 running = np.cumsum(water) / np.arange(1, 501)
5 fig, ax = plt.subplots()
6 ax.plot(running, color="#f9461c")
7 ax.axhline(0.75, color="#565a5c", linestyle="--")
8 ax.set(xlabel="toss", ylabel="proportion water", ylim=(0, 1))
```



Column Layout

Course Pattern

- Ask a data question
- Find the relevant table
- Check the units
- Make a readable view

Week	Focus
0	Setup and orientation
1	Blah blah
2	Blah blah blah
3	Blah blah blah blah

Auto-Animate



Auto-Animate

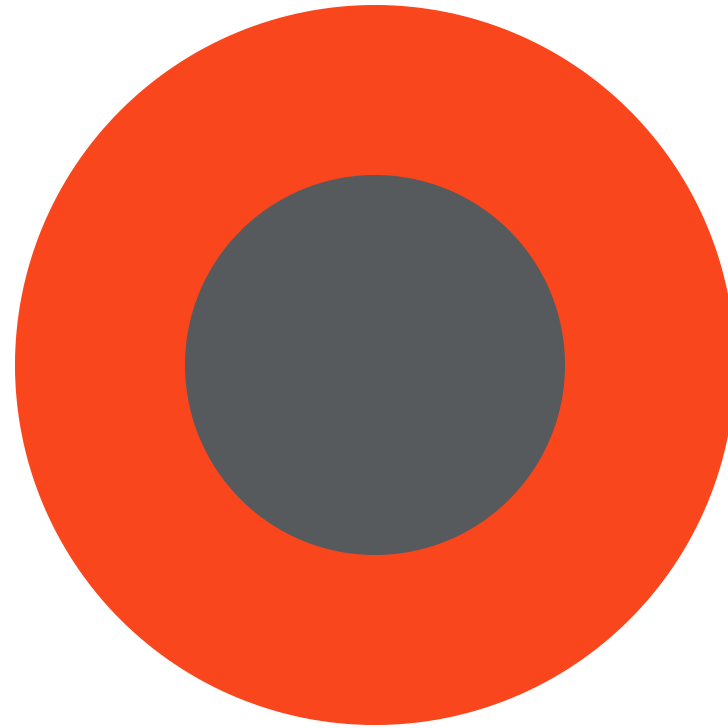
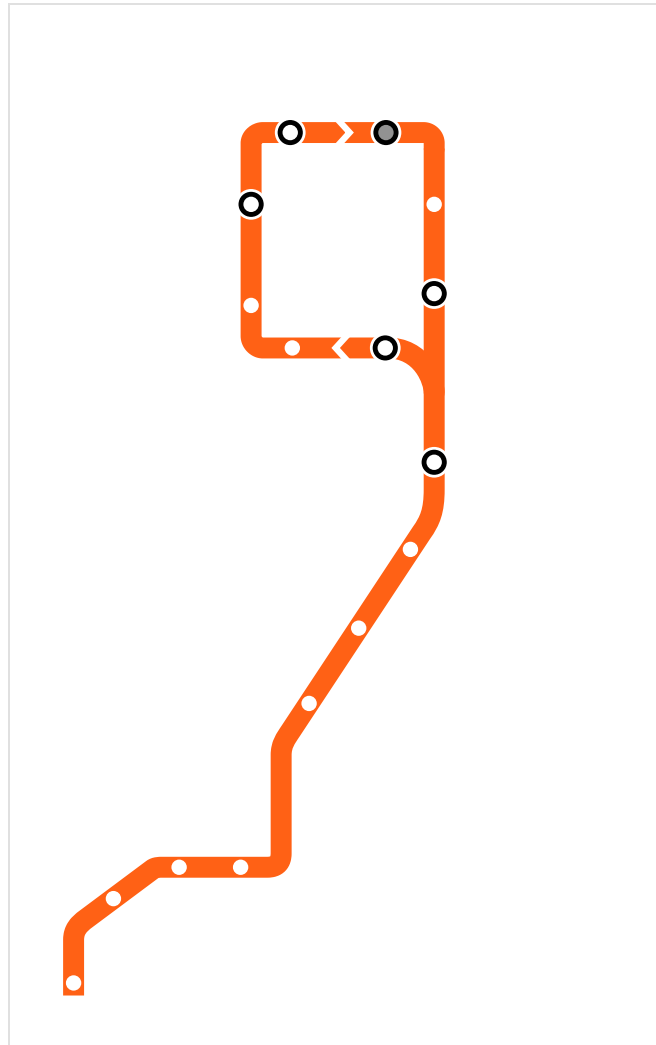


Image & Embed Slides

Image Frame Slide

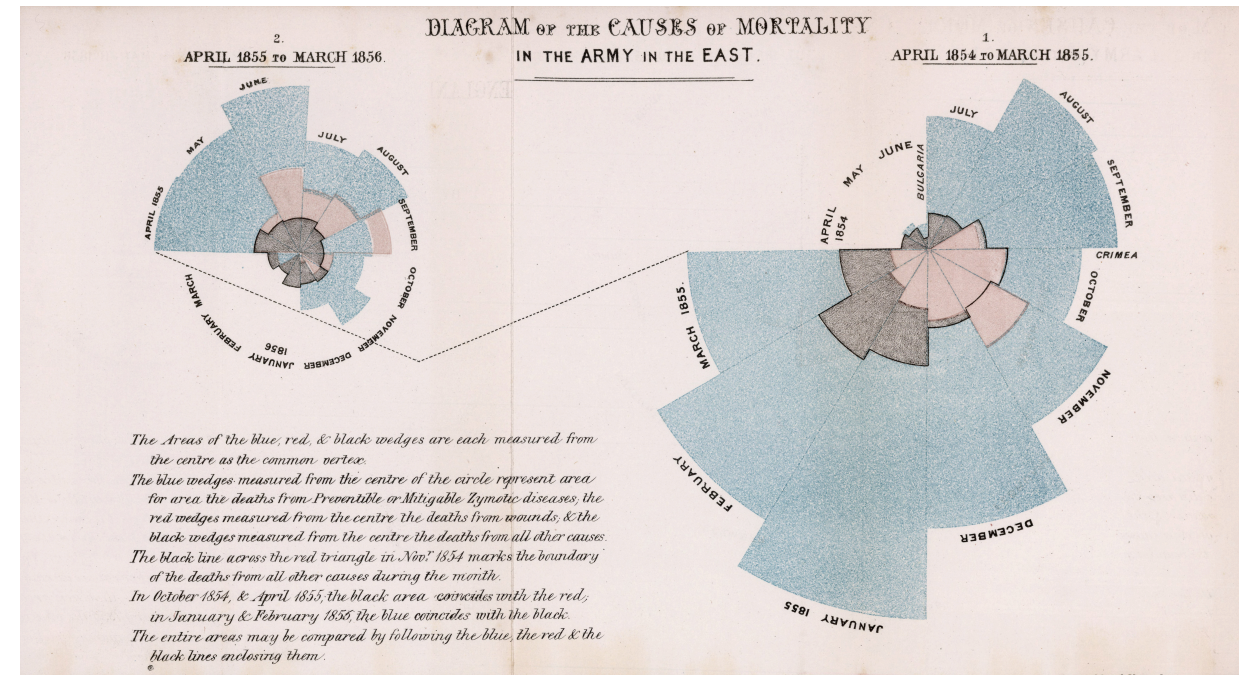


Florence Nightingale's Mortality Diagram (1858)

A **polar area chart** showing causes of mortality in the Crimean War:

- Blue: deaths from preventable disease
- Red: deaths from wounds
- Black: other causes

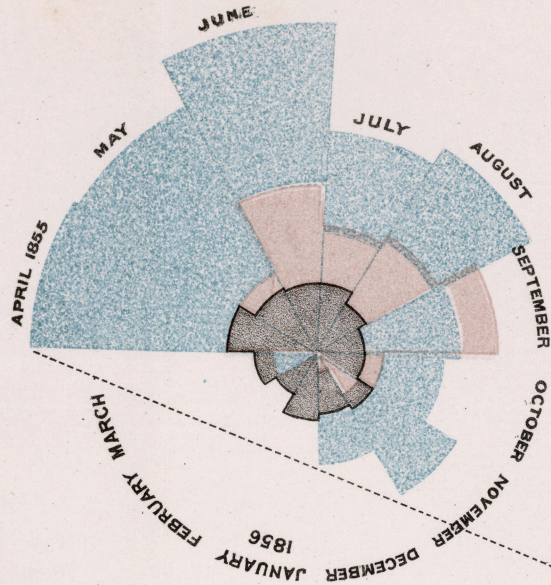
Used to advocate for sanitary reforms in military hospitals.



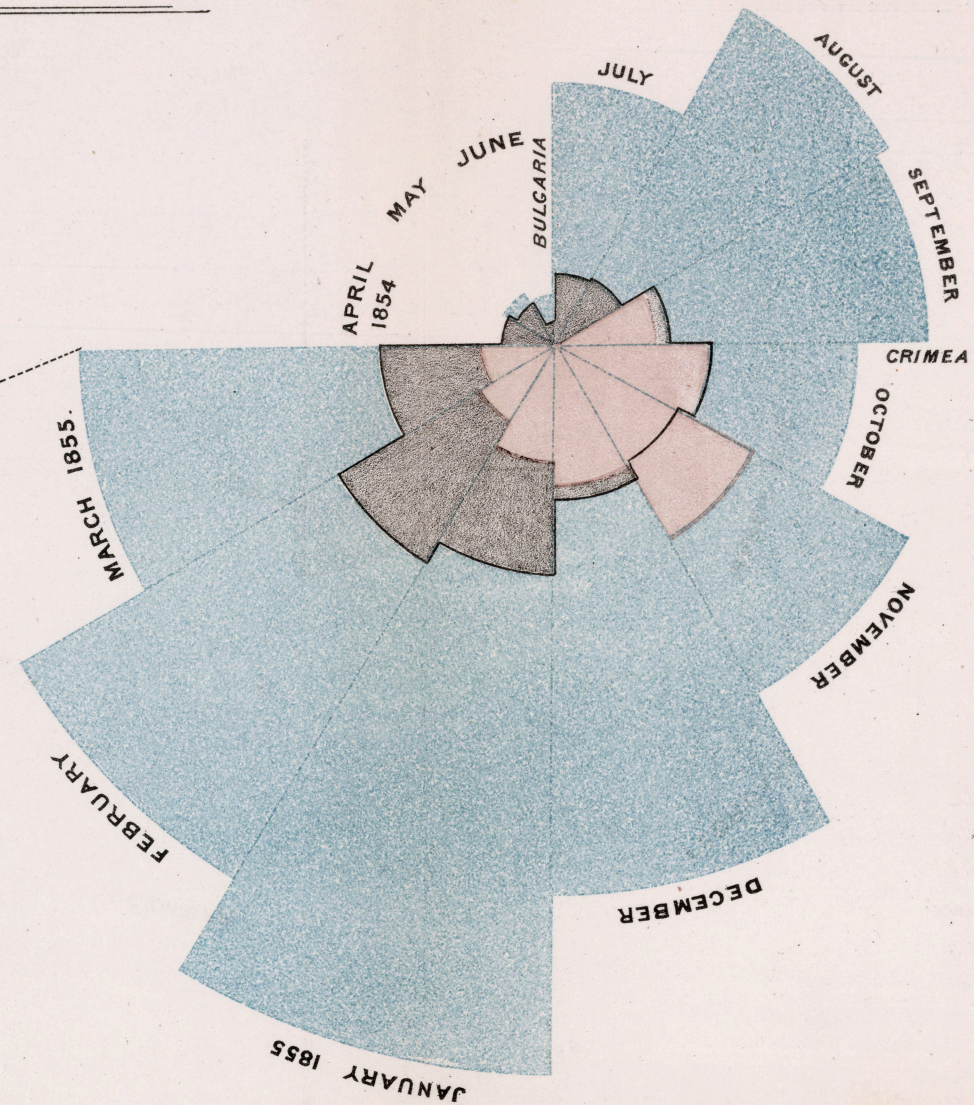
Florence Nightingale's Mortality Diagram (1858)

DIAGRAM OF THE CAUSES OF MORTALITY IN THE ARMY IN THE EAST.

2.
APRIL 1855 TO MARCH 1856.



1.
APRIL 1854 TO MARCH 1855.



The Areas of the blue, red, & black wedges are each measured from the centre as the common vertex.

The blue wedges measured from the centre of the circle represent area for area the deaths from Preventible or Mitigable Zymotic diseases; the red wedges measured from the centre the deaths from wounds; & the black wedges measured from the centre the deaths from all other causes.

The black line across the red triangle in Nov^r 1854 marks the boundary of the deaths from all other causes during the month.

In October 1854, & April 1855; the black area coincides with the red; in January & February 1855, the blue coincides with the black.

The entire areas may be compared by following the blue, the red & the black lines enclosing them.

Harriet Ann & Son, St. Martin's Lane.

follow-up slide for nightingale

- Lorem ipsum dolor sit amet,
- consectetur adipiscing elit
- Sed do eiusmod tempor incididunt
- ut labore et dolore magna

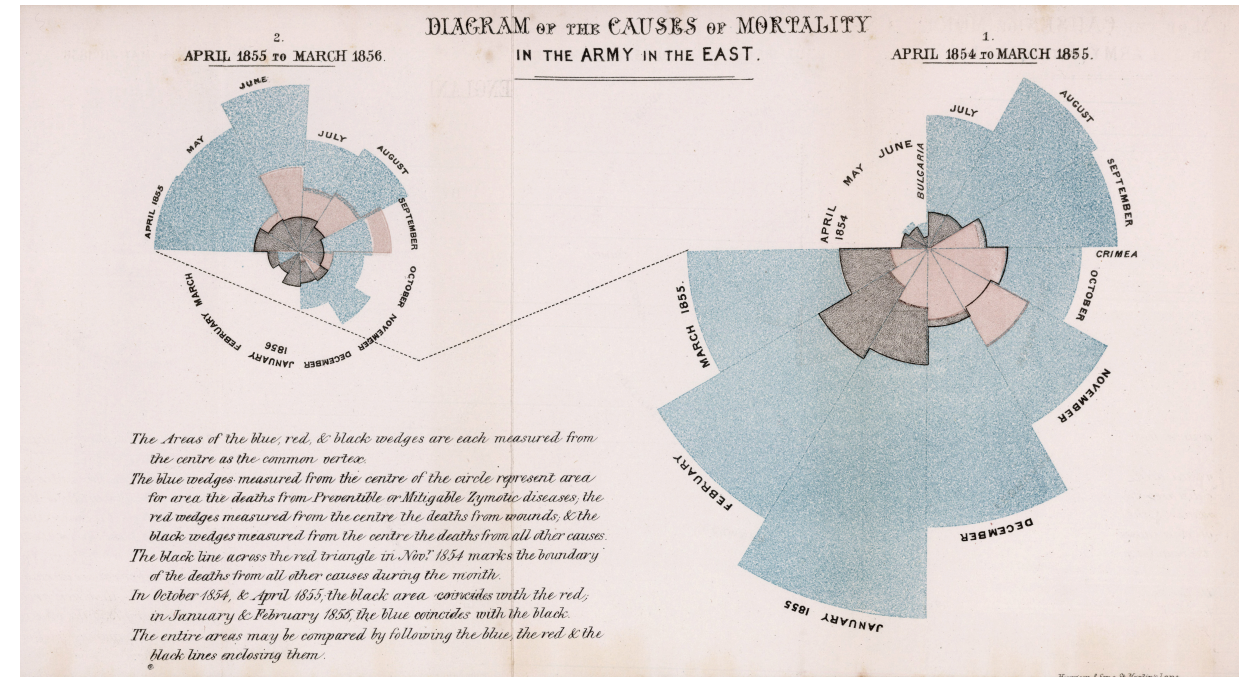
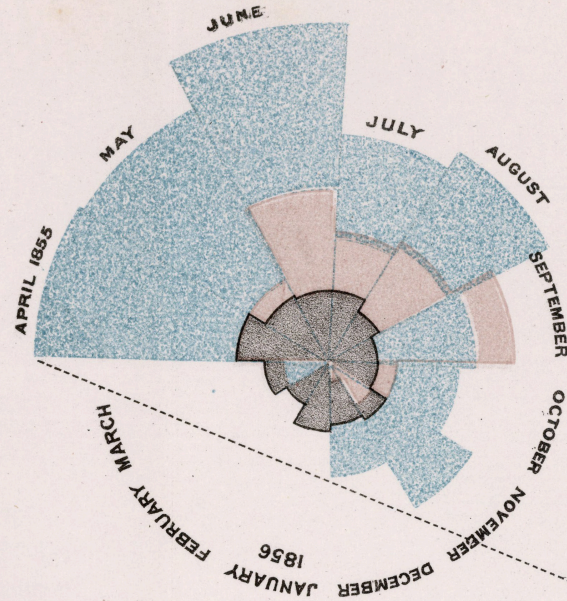
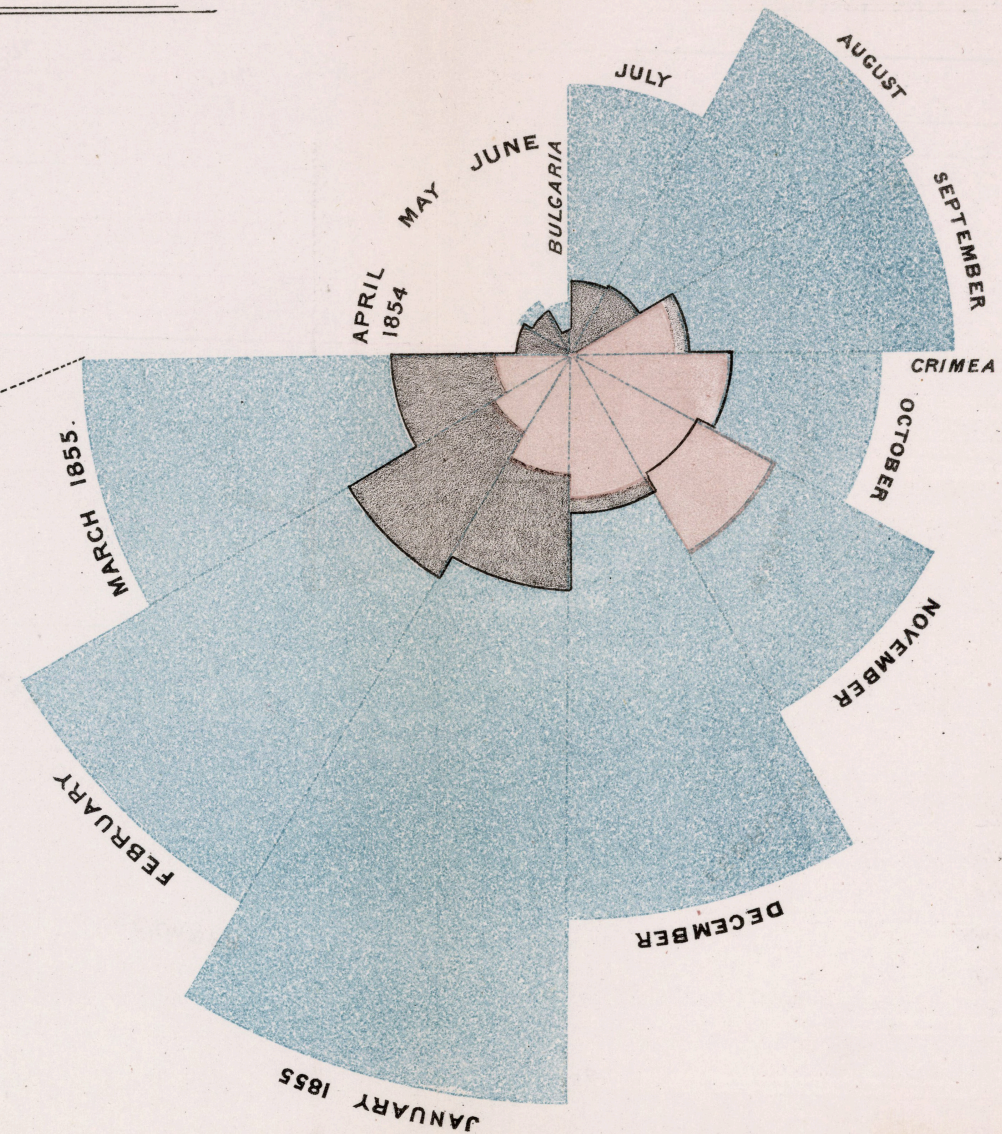


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University of Illinois Chicago



embedded web page: <https://www.uic.edu>

Sources

1. Nightingale, Florence. “Diagram of the causes of mortality in the army in the East.” *Notes on Matters Affecting the Health, Efficiency, and Hospital Administration of the British Army*, 1858. Image via Wikimedia Commons: <https://commons.wikimedia.org/wiki/File:Nightingale-mortality.jpg>. Public domain.
2. Slides developed using materials from [Elena Zheleva](#) and [Gonzalo Bello Lander](#), the Berkeley DS 100 team, Marine Carpuat, and Brian Ziebart.
3. Slide deck built with [Quarto](#) revealjs.
4. Last modified and compiled June 23, 12:35h Central (Chicago) time.
5. Title font is Big Shoulders; Body font is [Libre Franklin](#).